

Context-specific, directional, genetics-anchored drug-target nomination from genome-scale CD4+ T-cell CRISPRi Perturb-seq

Analysis report — computational target discovery

1. Summary

We re-analyzed a genome-scale CRISPRi Perturb-seq screen in primary human CD4+ T cells (Zhu, Dann, ... Pritchard & Marson 2025, bioRxiv 10.64898/2025.12.23.696273) to nominate drug targets for autoimmune and allergic disease. Our angle is deliberately differentiated from the preprint's own target list along three axes:

1. **Directionality** — we separate targets whose knockdown *lowers* a pathogenic pro-inflammatory program (**drivers** → **antagonize**) from those whose knockdown *raises* it (**brakes** → **agonize**), because the two imply opposite therapeutic modalities (inhibit vs. potentiate).
2. **Human-genetics anchoring** — every nomination is scored against Open Targets autoimmune/allergic disease genetics and gnomAD loss-of-function constraint.
3. **Druggability + clinical-novelty filter** — we attach a tractability tier and suggested modality to each target, and flag which nominations are *not yet drugged*, surfacing hypotheses that are both actionable and new.

From **1,923** quality-filtered causal candidates (**1,290 driver**→**antagonize**, **633 brake**→**agonize**), the integration recovers known immune drug targets at the very top (IL4R #1, IL2RA #2, STAT3 #3, IL6R #7 — all approved or clinical-stage), which validates the scoring, and it nominates **86 novel-priority targets** (novel/undrugged × strong-or-moderate genetics × druggable) — **48 to antagonize**, **38 to agonize**.

Nature of the claims. These are *computational* hypotheses built by triangulating orthogonal evidence (causal perturbation effect + human genetics + druggability + clinical novelty). No wet-lab validation was performed here. Their credibility rests on the convergence of independent evidence layers and on the recovery of established targets as positive controls, not on any single statistic.

2. Data and the pathogenic program

Input. `GWCD4i.DE_stats.h5ad` — per-perturbation differential-expression statistics (z-scores of downstream readout genes) for genome-scale CRISPRi across three conditions: **Rest**, **Stim 8 h**, **Stim 48 h**. The DE matrix is 33,983 (perturbation × condition) rows × 10,282 readout genes.

Pro-inflammatory program signature. We defined a Th1/pathogenic-effector program as a signed gene set: **36 pro-inflammatory genes (+1)** (e.g. *IFNG*, *TBX21*, *STAT1*, *STAT4*, *CXCR3*) and **12 regulatory/anti-inflammatory genes (-1)** (e.g. *IL10*, *FOXP3*, *CTLA4*). 48 of 50 intended genes were measured in the readout (*IL17A*, *RORA* absent). A per-perturbation **program score** is the mean of (signed weight × readout z-score) over signature genes; negative = program suppressed, positive = program elevated.

(Figure: `program_signature_heatmap.png` — signature-gene readout across positive-control perturbations.)

3. Directional, context-specific scoring

For each perturbation × condition we computed the program score and calibrated it against an **empirical null**: 500 random signed gene sets matched in size (36 up / 12 down) drawn from non-signature genes, giving an empirical z

(emp_z) and a BH-FDR per condition.

- **Direction.** A perturbation whose knockdown drives the program *down* is a **driver** → **antagonize** target; one that drives it *up* is a **brake** → **agonize** target.
- **Context specificity.** We record the peak condition (Rest / Stim 8 h / Stim 48 h) and a context-specificity index (0 = uniform across conditions, 1 = single-condition), because a target that acts only in the activated state has a different therapeutic window than a constitutive one.

Quality gate. A candidate must show (a) achieved knockdown (peak_kd < -1.0 on-target) and (b) a program hit (peak_emp_fdr < 0.05 and |peak_score| > 0.2). **1,923** perturbations pass.

Positive controls behaved as expected: knockdown of drivers *TBX21*, *BATF*, *STAT3*, *IRF4* lowered the program; knockdown of brakes *STAT6*, *IL4R*, *INPP5D* raised it.

(Figure: *perturbation_directionality.png* — volcano/directionality map of all candidates.)

4. Human-genetics support

Each candidate's Ensembl ID was queried against **Open Targets** for autoimmune/allergic disease associations and against gnomAD for LoF constraint. We assigned a **genetics tier** (strong / moderate / weak / none) combining autoimmune association score and LoF-constraint. **188** candidates reach the *strong* tier; **125** of those are driver→antagonize.

(Figure: *genetics_support.png* — genetics score vs. causal effect, colored by direction.)

5. Druggability and tractability

We mapped each target to a protein class and assigned a **suggested modality** with a priority that respects biology:

- secreted cytokine → **biologic (secreted)**;
- cytokine/catalytic receptor (cell-surface) → **antibody**;
- kinase / enzyme / GPCR / ion channel / nuclear receptor / transporter → **small molecule**;
- otherwise small-molecule-ligandable/pocket → **small molecule**; surface antigen → **antibody**;
- transcription factor with no pocket → **hard (degrader/PPI)**.

A **tractability tier** (clinical precedent → discovery precedent → predicted tractable → difficult) captures how far modality development has progressed. Among strong-genetics antagonize targets, **67** are druggable; by direction the antagonize set splits roughly into ~210 small-molecule, ~34 antibody, and ~106 hard-TF targets.

(Figure: *druggability_landscape.png* — modality × tractability landscape.)

6. Clinical novelty

Using Open Targets known-drug/clinical-candidate records we classified each target into: **novel/undrugged**, **preclinical/unknown**, **clinical-stage drug**, **approved drug**. Among the deep-annotated top-200 antagonize set, **47** targets are *novel/undrugged with strong genetics* — the differentiated core of this analysis. *STAT3* correctly re-surfaced as *clinical-stage*, confirming that the novelty filter recovers known-drug status.

(Figure: *novelty_breakdown.png* — novelty × genetics stacked bar + novel-undrugged target lollipop.)

7. Integrated ranking

We combined four evidence components, each normalized to [0,1], into a single `integrated_score`:

Component	Weight	Source
Causal strength	0.34	rank-normalized empirical z at peak condition
Human genetics	0.30	Open Targets autoimmune association + LoF-constraint bonus
Druggability	0.22	tractability tier (hard-TF floored)
Clinical novelty	0.14	undrugged > preclinical > clinical > approved

Directionality is preserved as an *annotation*, not folded into the score, so the user can filter to antagonize or agonize as the therapeutic question demands.

Positive-control recovery (validation):

Target	Rank	Score	Status
IL4R	1	0.87	approved (dupilumab; agonize-brake control)
IL2RA	2	0.84	approved (basiliximab)
STAT3	3	0.83	clinical-stage
IL6R	7	0.79	approved (tocilizumab)
TBX21	20	0.75	known driver
BATF	39	0.70	known driver

The clustering of approved/clinical immune targets at the top of an unsupervised integrated score is the primary internal validation of the pipeline.

(Figure: `top_targets_evidence_matrix.png` — evidence heatmap × composite score × modality/disease for the top nominations.)

8. Top novel nominations (deep dive)

Six novel-priority nominations spanning both directions, diverse immune diseases, and tractable modalities:

Target	Direction	Rank	Peak	Genetics (AI)	Disease anchor	Modality
STAT5B	antagonize	15	Stim 48 h	0.81	GH-insensitivity w/ immune dysregulation	small molecule
CTSH	antagonize	31	Stim 8 h	0.89	type 1 diabetes	small molecule

NEK6	antagonize	45	Stim 8 h	0.81	asthma	small molecule (kinase)
PTPN2	agonize	13	Rest	0.89	rheumatoid arthritis	small molecule
SIK2	agonize	19	Rest	0.75	asthma	small molecule (kinase)
STAT6	agonize	11	Stim 48 h	0.81	hyper-IgE syndrome 6	small molecule

Each shows a distinct context-specific trajectory ([top_target_vignettes.png](#)): the antagonize drivers *STAT5B/CTSH/NEK6* act on activation (Stim 8–48 h), while the brakes *PTPN2/SIK2* act at Rest and *STAT6* emerges late (Stim 48 h). The two kinases (NEK6, SIK2) and the phosphatase (PTPN2) are especially attractive small-molecule handles; CTSH (type-1-diabetes genetics) and STAT6 (already a validated allergy axis via its pathway) anchor to strong disease genetics.

Caveats specific to the deep-dive set. Some disease anchors are Mendelian immune-dysregulation syndromes (e.g. STAT5B, STAT6 hyper-IgE) rather than common-variant GWAS hits; these support a *causal role in immune regulation* but the direction of a common-disease effect should be confirmed against common-variant data before prioritization. Directionality here is defined relative to our program signature, not to clinical outcome.

9. Limitations

- **No wet-lab validation.** All nominations are computational hypotheses; the intended next step is arrayed CRISPRi/knockdown or small-molecule validation of program effect in primary CD4+ T cells.
- **Signature dependence.** The program score depends on the chosen 48-gene signature; *IL17A/RORA* were unmeasured, so a Th17-skewed program is under-represented.
- **Genetics coverage.** Open Targets associations mix common-variant GWAS and Mendelian evidence; tier assignment does not distinguish direction of genetic effect.
- **Druggability heuristics.** Modality assignment is rule-based from protein class; it approximates but does not replace structural tractability assessment.
- **Single cell type / stimulus axis.** Findings are specific to CD4+ T cells under anti-CD3/CD28-type stimulation and may not transfer to other immune compartments.

10. Deliverables

File	Contents
target_shortlist.csv	Ranked, evidence-annotated shortlist (1,923 candidates × 33 columns)
top_targets_evidence_matrix.png	Evidence heatmap + composite + modality/disease for top nominations
perturbation_directionality.png	Directional context-specific scoring map
genetics_support.png	Human-genetics support layer
druggability_landscape.png	Druggability / tractability landscape
novelty_breakdown.png	Clinical-novelty × genetics breakdown

<code>top_target_vignettes.png</code>	Deep-dive trajectories for 6 novel nominations
<code>deep_dive_vignettes.csv</code>	Full evidence rows for the 6 deep-dive targets
<code>target_nomination_pipeline.ipynb</code>	Reproducible end-to-end pipeline

*Data source: Zhu, Dann, ... Pritchard & Marson (2025), bioRxiv 10.64898/2025.12.23.696273.
Genetics/druggability/novelty via Open Targets and gnomAD.*